Problem Set 3: Two-Period New Keynesian Model

Macroeconomics II (Spring 2026)
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Problem 1 (Questions 1–20): IS / Euler block

1 **Timeline in two-period NK. Which expectation operator is nontrivial?** B

(A) $\mathbb{E}_1[\cdot]$ about period 2 variables

(B) $\mathbb{E}_0[\cdot]$ about period 1 variables

(C) $\mathbb{E}_{-1}[\cdot]$ about period 0 variables

(D) No expectations appear in a two-period model

Explanations.

(A) Wrong: no period 2 exists.

(B) Correct: only expectations from 0 to 1 matter.

(C) Wrong: not relevant here.

(D) Wrong: expectations can matter for period 1.

2 **Fisher relation (ex-ante real rate). Which is correct?** C

(A) $r_0 = i_0 + \mathbb{E}_0\pi_1$

(B) $r_0 = \mathbb{E}_0\pi_1 - i_0$

(C) $r_0 = i_0 - \mathbb{E}_0\pi_1$

(D) $r_0 = i_0 - \pi_0$

Explanations.

(A) Wrong sign.

(B) Wrong sign.

(C) Correct.

(D) Wrong: uses current instead of expected inflation.

③ **NK IS (Euler) equation: pick the standard form.** A

(A) $x_0 = \mathbb{E}_0 x_1 - \frac{1}{\sigma}(i_0 - \mathbb{E}_0\pi_1 - r_0^n)$

(B) $x_0 = \mathbb{E}_0 x_1 + \frac{1}{\sigma}(i_0 - \mathbb{E}_0\pi_1 - r_0^n)$

(C) $x_0 = \frac{1}{\sigma}(i_0 - \mathbb{E}_0\pi_1 - r_0^n)$

(D) $x_0 = \mathbb{E}_0 x_1 - \sigma(i_0 - \mathbb{E}_0\pi_1 - r_0^n)$

Explanations.

(A) Correct.

(B) Wrong sign.

(C) Missing expectation.

(D) Wrong coefficient.

④ **Meaning of σ in NK IS.** C

(A) Policy response to inflation

(B) Price rigidity parameter

(C) Inverse intertemporal elasticity of substitution

(D) Slope of Phillips curve

Explanations.

(A) Wrong.

(B) Wrong.

(C) Correct.

(D) Wrong.

⑤ **If $x_1 = 0$ for sure, NK IS implies:** B

(A) $x_0 = \mathbb{E}_0 x_1$

(B) $x_0 = -\frac{1}{\sigma}(i_0 - \mathbb{E}_0\pi_1 - r_0^n)$

(C) $x_0 = +\frac{1}{\sigma}(i_0 - \mathbb{E}_0\pi_1 - r_0^n)$

(D) $x_0 = 0$ always

Explanations.

(A) Wrong.

(B) Correct.

(C) Wrong sign.

(D) Wrong.

6 **Numerical IS.** $\sigma = 2$, $r_0^n = 1\%$, $i_0 = 3\%$, $\mathbb{E}_0\pi_1 = 1\%$, $x_1 = 0$. **Then** $x_0 =$ __C__

(A) $+0.5\%$

(B) 0

(C) -0.5%

(D) -1%

Explanations.

(A) Wrong sign.

(B) Wrong.

(C) Correct.

(D) Wrong: forgot $1/\sigma$.

7 **Natural real rate** r_0^n **is:** __D__

(A) The nominal policy rate

(B) Inflation target

(C) Real rate consistent with $x_0 = 0$ under sticky prices

(D) Real rate consistent with $x_0 = 0$ under flexible prices

Explanations.

(A) Wrong.

(B) Wrong.

(C) Close but not definition.

(D) Correct.

(8) **Holding i_0 fixed, higher $\mathbb{E}_0\pi_1$ implies:** __B__

(A) Higher r_0 and lower x_0

(B) Lower r_0 and higher x_0

(C) Lower r_0 and lower x_0

(D) No effect on x_0

Explanations.

(A) Wrong.

(B) Correct.

(C) Wrong.

(D) Wrong.

(9) **ZLB constraint is:** __C__

(A) $r_0 \geq 0$

(B) $\pi_0 \geq 0$

(C) $i_0 \geq 0$

(D) $x_0 \geq 0$

Explanations.

(A) Wrong.

(B) Wrong.

(C) Correct.

(D) Wrong.

(10) **At ZLB, if expected inflation falls, then:** __A__

(A) Real rate rises and demand falls

(B) Real rate falls and demand rises

(C) Nominal rate rises automatically

(D) NKPC shifts down only

Explanations.

(A) Correct.

(B) Wrong.

(C) Wrong.

(D) Wrong.

11. **NK Phillips curve (NKPC) form is:** B

(A) $\pi_0 = \pi_{-1} + \kappa x_0 + u_0$

(B) $\pi_0 = \beta \mathbb{E}_0 \pi_1 + \kappa x_0 + u_0$

(C) $\pi_0 = \beta \pi_1 - \kappa x_0 + u_0$

(D) $\pi_0 = \kappa x_0 + u_0$

Explanations.

(A) Backward-looking.

(B) Correct.

(C) Wrong.

(D) Missing expectations.

12. **Interpretation of β in NKPC:** A

(A) Discount factor weight on expected inflation

(B) Policy response to inflation

(C) IS slope parameter

(D) Cost-push shock size

Explanations.

(A) Correct.

(B) Wrong.

(C) Wrong.

(D) Wrong.

(13) **Interpretation of u_0 in NKPC:** A

(A) Cost-push/markup shock

(B) Demand shock

(C) Natural rate

(D) Nominal rate

Explanations.

(A) Correct.

(B) Wrong.

(C) Wrong.

(D) Wrong.

(14) **If $\mathbb{E}_0\pi_1 = 0$, NKPC becomes:** A

(A) $\pi_0 = \kappa x_0 + u_0$

(B) $\pi_0 = \beta + \kappa x_0 + u_0$

(C) $\pi_0 = \beta\pi_0 + \kappa x_0 + u_0$

(D) $\pi_0 = u_0$

Explanations.

(A) Correct.

(B) Wrong.

(C) Wrong.

(D) Wrong.

(15) **Taylor rule (simple) is:** A

(A) $i_0 = \bar{i} + \phi_\pi \pi_0 + \phi_x x_0$

(B) $i_0 = r_0^n$

(C) $i_0 = \pi_0 - \phi_\pi x_0$

(D) $i_0 = \mathbb{E}_0\pi_1$

Explanations.

(A) Correct.

(B) Wrong.

(C) Wrong.

(D) Wrong.

(16) **Taylor principle typically requires:** C

(A) $\phi_\pi < 1$

(B) $\phi_\pi = 1$

(C) $\phi_\pi > 1$

(D) $\phi_\pi < 0$

Explanations.

(A) Wrong.

(B) Boundary.

(C) Correct.

(D) Wrong.

(17) **Determinacy intuition: when inflation rises, real rate should:** A

(A) Rise (tighten)

(B) Fall (loosen)

(C) Stay constant

(D) Be undefined

Explanations.

(A) Correct.

(B) Wrong.

(C) Wrong.

(D) Wrong.

(18) **Demand shock (IS) tends to move:** A

(A) $x \uparrow, \pi \uparrow$

(B) $x \downarrow, \pi \uparrow$

(C) $x \uparrow, \pi \downarrow$

(D) $x \downarrow, \pi \downarrow$

Explanations.

(A) Correct.

(B) Cost-push.

(C) Wrong.

(D) Wrong.

(19) **Cost-push shock ($u_0 > 0$) tends to move:** B

(A) $x \uparrow, \pi \uparrow$

(B) $x \downarrow, \pi \uparrow$

(C) $x \uparrow, \pi \downarrow$

(D) $x \downarrow, \pi \downarrow$

Explanations.

(A) Wrong.

(B) Correct.

(C) Wrong.

(D) Wrong.

(20) **Larger κ means (holding x fixed):** B

(A) Inflation responds less

(B) Inflation responds more

(C) No change

(D) Inflation becomes backward-looking

Explanations.

(A) Wrong.

(B) Correct.

(C) Wrong.

(D) Wrong.

Problem 2 (Questions 21–40): NKPC + Taylor rule

21 **Numerical NKPC:** $\beta = 1$, $\mathbb{E}_0\pi_1 = 1\%$, $\kappa = 0.5$, $x_0 = 2\%$, $u_0 = 0$. **Then** $\pi_0 =$ B

(A) 1%

(B) 2%

(C) 2%

(D) 0.5%

Explanations.

(A) Wrong: forgot gap term.

(B) Correct: $1\% + 0.5imes2\% = 2\%$.

(C) Duplicate value; treated as distractor.

(D) Wrong.

22 **Numerical Taylor:** $ari = 1\%$, $\phi_\pi = 1.5$, $\phi_x = 0$, $\pi_0 = 2\%$. **Then** $i_0 =$ C

(A) 2%

(B) 3%

(C) 4%

(D) 1%

Explanations.

(A) Wrong.

(B) Wrong.

(C) Correct.

(D) Wrong.

(23) **Compute real rate: $i_0 = 4\%$, $\mathbb{E}_0\pi_1 = 1\%$. Then $r_0 =$** __A__

(A) 3%

(B) 5%

(C) -3%

(D) 1%

Explanations.

(A) Correct.

(B) Wrong.

(C) Wrong.

(D) Wrong.

(24) **If CB raises i_0 holding expectations fixed, IS implies x_0:** __B__

(A) Increases

(B) Decreases

(C) Unchanged

(D) Undefined

Explanations.

(A) Wrong.

(B) Correct.

(C) Wrong.

(D) Wrong.

(25) **If terminal conditions are $x_1 = 0$ and $\pi_1 = 0$, then:** __A__

(A) $\mathbb{E}_0 x_1 = \mathbb{E}_0 \pi_1 = 0$

(B) Expectations become larger

(C) Expectations are indeterminate

(D) Expectations equal current values

Explanations.

(A) Correct.

(B) Wrong.

(C) Wrong.

(D) Wrong.

26 **If ϕ_x increases, policy responds more to:** B

(A) Inflation only

(B) Output gap (stabilization)

(C) Expected inflation only

(D) Natural rate only

Explanations.

(A) Wrong.

(B) Correct.

(C) Wrong.

(D) Wrong.

27 **If $\phi_\pi = 0$ (peg), higher inflation tends to:** B

(A) Raise real rate

(B) Lower real rate

(C) Leave real rate unchanged

(D) Stabilize inflation

Explanations.

(A) Wrong.
(B) Correct.
(C) Wrong.
(D) Wrong.

28 **With $\phi_\pi > 1$, inflation rise tends to:** B

(A) Reduce real rate
(B) Increase real rate
(C) Not change real rate
(D) Set $x_0 = 0$ always

Explanations.

(A) Wrong.
(B) Correct.
(C) Wrong.
(D) Wrong.

29 **If expected inflation rises but rule reacts to current π_0 only, i_0 changes:** B

(A) Immediately due to expectations
(B) Only if π_0 changes
(C) One-for-one always
(D) Never

Explanations.

(A) Wrong.
(B) Correct.
(C) Wrong.
(D) Wrong.

30 **Which blocks are forward-looking?** C

(A) IS only

(B) NKPC only

(C) Both IS and NKPC

(D) Neither

Explanations.

(A) Wrong.

(B) Wrong.

(C) Correct.

(D) Wrong.

31 **If $u_0 > 0$ and CB wants lower π_0, it chooses:** B

(A) $x_0 > 0$

(B) $x_0 < 0$

(C) $x_0 = 0$

(D) Any x_0

Explanations.

(A) Wrong.

(B) Correct.

(C) Wrong.

(D) Wrong.

32 **If β rises toward 1, NKPC becomes:** B

(A) Less forward-looking

(B) More forward-looking

(C) Backward-looking

(D) Unrelated to expectations

Explanations.

(A) Wrong.
(B) Correct.
(C) Wrong.
(D) Wrong.

33. **If $\mathbb{E}_0\pi_1$ rises holding x_0 fixed, π_0 rises by:** B

(A) 0
(B) β times the change
(C) $(1-\beta)$ times the change
(D) κ times the change

Explanations.

(A) Wrong.
(B) Correct.
(C) Wrong.
(D) Wrong.

34. **If $\kappa = 0$, then inflation is:** A

(A) Independent of x_0
(B) More sensitive to x_0
(C) Always equal to target
(D) Always zero

Explanations.

(A) Correct.
(B) Wrong.
(C) Wrong.
(D) Wrong.

35. **If σ is very large, IS response to real-rate gap is:** B

(A) Strong

(B) Weak

(C) Zero always

(D) Opposite sign

Explanations.

(A) Wrong.

(B) Correct.

(C) Wrong.

(D) Wrong.

36 **If r_0^n increases holding i_0 and expectations fixed, then x_0:** B

(A) Falls

(B) Rises

(C) Unchanged

(D) Depends only on NKPC

Explanations.

(A) Wrong.

(B) Correct.

(C) Wrong.

(D) Wrong.

37 **If CB sets $i_0 = r_0^n + \mathbb{E}_0\pi_1$ and $x_1 = 0$, then $x_0 =$** B

(A) > 0

(B) $= 0$

(C) < 0

(D) Undefined

Explanations.

(A) Wrong.
(B) Correct.
(C) Wrong.
(D) Wrong.

(38) **If $i_0 = 0$ and $\mathbb{E}_0\pi_1 = -2\%$, then $r_0 =$** __C__

(A) -2%
(B) 0%
(C) 2%
(D) -4%

Explanations.
(A) Wrong.
(B) Wrong.
(C) Correct.
(D) Wrong.

(39) **A higher ϕ_π tends to make equilibrium inflation (given shocks):** __B__

(A) More sensitive
(B) Less sensitive
(C) Unchanged
(D) Always zero

Explanations.
(A) Wrong.
(B) Correct.
(C) Wrong.
(D) Wrong.

(40) **Anchored expectations roughly means:** __A__

(A) $\mathbb{E}_0\pi_1$ stays near target

(B) π_0 always zero

(C) x_0 always zero

(D) i_0 never changes

Explanations.

(A) Correct.

(B) Wrong.

(C) Wrong.

(D) Wrong.